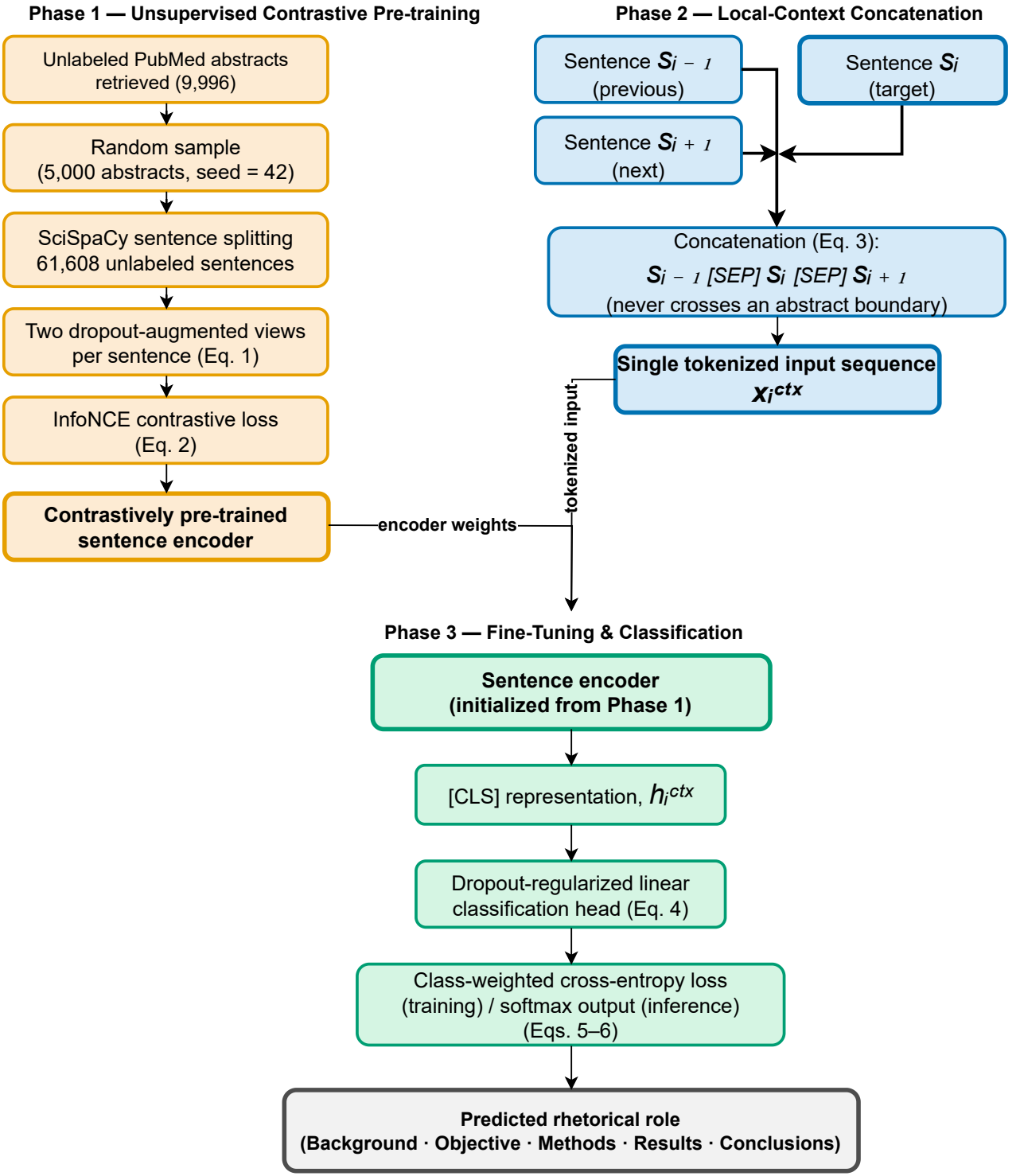

Phase 1 — Unsupervised Contrastive Pre-training
Unlabeled PubMed abstracts retrieved (9,996)
Random sample (5,000 abstracts, seed = 42)
SciSpaCy sentence splitting 61,608 unlabeled sentences
Two dropout-augmented views per sentence (Eq. 1)
InfoNCE contrastive loss (Eq. 2)
Contrastively pre-trained sentence encoder
Phase 2 — Local-Context Concatenation
Sentence S_{i-1} (previous)
Sentence S_i (target)
Sentence S_{i+1} (next)
Concatenation (Eq. 3):
S_{i-1} [SEP] S_i [SEP] S_{i+1}
(never crosses an abstract boundary)
Single tokenized input sequence x_i^{ctx}
encoder weights
tokenized input
Phase 3 — Fine-Tuning & Classification
Sentence encoder (initialized from Phase 1)
[CLS] representation, h_i^{ctx}
Dropout-regularized linear classification head (Eq. 4)
Class-weighted cross-entropy loss (training) / softmax output (inference) (Eqs. 5–6)
Predicted rhetorical role
(Background · Objective · Methods · Results · Conclusions)